Project Report: Output Characteristics & Analysis of N-Channel JFET

1. Title & Aim

- **Project Title:** Simulation and Analysis of N-Channel JFET Characteristics
- **Aim:**
    - To simulate and obtain the Drain/Output characteristics (I_D vs. V_DS) of an N-Channel JFET for varying Gate-to-Source voltages (V_GS).
    - To determine the dynamic drain resistance (r_d), transconductance (g_m), and pinch-off voltage (V_P).
    - To demonstrate and explain why a JFET behaves as a voltage-controlled current source.
    - To demonstrate and explain why a JFET possesses high input impedance.

2. Circuit Diagram & Simulation Setup

Components Required

- **N-Channel JFET:** 2N3819 (or standard N-JFET model)
- **Variable DC Power Supplies:** V_DD (0–15V) and V_GG (0–3V negative bias)
- **Voltmeters:** V_DS measurement across Drain-Source, V_GS measurement across Gate-Source
- **Ammeters:** I_D (Drain Current in mA), I_G (Gate Current in µA/nA)
- **Resistors:** R_D = 100 Ω, R_G = 1 MΩ

Circuit Connections

- Connect the Gate terminal of the N-JFET to the negative terminal of V_GG so that the PN junction is reverse-biased.
- Connect the Source terminal to ground.
- Connect the Drain terminal through a resistor R_D to the positive terminal of V_DD.

None

```
        +V_DD (0 to 15V)
          |
         [R_D = 100 Ohm]
```

```
         |
         +---- (Ammeter I_D) ----+
                                 |

                                D | |--- Drain
|---|2N3819)
    (-0V to -3V)                S | |--- Source
                                 |
                                GND
```

3. Observation Table & Readings

Table 1: Output Characteristics Data (I_D vs. V_DS for different V_GS)

V_DS (V)	I_D (mA) at V_GS = 0 V	I_D (mA) at V_GS = -0.5 V	I_D (mA) at V_GS = -1.0 V	I_D (mA) at V_GS = -2.0 V
0.0	0.00	0.00	0.00	0.00
0.5	1.80	1.30	0.85	0.25
1.0	3.50	2.50	1.60	0.45
1.5	5.10	3.60	2.20	0.60
2.0 (V_P)	6.20	4.30	2.60	0.70
4.0	6.45	4.45	2.70	0.72
6.0	6.50	4.50	2.75	0.73
8.0	6.52	4.52	2.76	0.74
10.0	6.55	4.55	2.78	0.75

Table 2: Input Gate Current Data (I_G vs. V_GS)

Gate Voltage V_GS (V)	Gate Leakage Current I_G	Input Impedance Z_in = |V_GS / I_G|
0.0 V	0.01 nA	Infinite

Gate Voltage V_GS (V)	Gate Leakage Current I_G	Input Impedance Z_in = |V_GS / I_G|
-1.0 V	0.05 nA	20 GΩ
-2.0 V	0.10 nA	20 GΩ
-3.0 V	0.15 nA	20 GΩ

4. Screenshot / Waveform Layout

Circuit Schematic Setup

Displays the N-JFET connected with multimeters measuring V_DS, V_GS, I_D, and I_G.

Output Characteristics Plot (I_D vs V_DS)

```
None
  I_D (mA) ^
         8 |     ---------------------------- V_GS = 0V (I_DSS = 6.5 mA)
         6 |---/------------------------------ V_GS = -0.5V
         4 |--/------------------------------- V_GS = -1.0V
         2 |-/-------------------------------- V_GS = -2.0V
         0 +----------------------------------> V_DS (V)
           0   2(V_P)  4    6    8    10   12
           |<Ohmic>|<--- Saturation Region --->|
                    (Constant Current Region)
```

5. Theoretical Analysis & Questions Answered

(i) Why JFET is a Voltage-Controlled Current Source

- **Working Principle:**
  - The drain current I_D in the saturation region is defined by Shockley's Equation:
    `I_D = I_DSS * (1 - V_GS / V_P)^2`
  - Where I_DSS is the maximum saturation drain current (at V_GS = 0 V) and V_P is the pinch-off voltage.
- **Inference:**

- Varying the input voltage V_{GS} modulates the width of the depletion layer in the channel, directly controlling the magnitude of the output current I_D in the saturation region.
- Because negligible input current flows ($I_G \approx 0$), the input parameter controlling the output is strictly voltage (V_{GS}), establishing the JFET as a Voltage-Controlled Current Source (VCCS).

(ii) Why JFET has High Input Impedance

- **Reverse-Biased Gate-Source Junction:**
  - Under normal operating conditions, the Gate-Source PN junction is maintained in reverse bias ($V_{GS} \leq 0$ V).
  - A reverse-biased junction supports only minor minority carrier drift, limiting leakage current to nanoamperes or picoamperes.
- **Mathematical Proof from Data:**
  - $Z_{in} = \Delta V_{GS} / \Delta I_G = 1.0 \text{ V} / 0.05 \text{ nA} = 20 \text{ G}\Omega$
- **Inference:**
  - Because $I_G \approx 0$, the input impedance of a JFET is exceptionally high (typically 10^8 to 10^11 Ω), minimizing signal loading effects when compared to BJT configurations.

6. Conclusion

- **Ohmic Region:** For $V_{DS} < V_P$, the channel behaves as a linear resistance where I_D increases proportionally with V_{DS}.
- **Saturation Region:** For $V_{DS} \geq V_P$, pinch-off occurs at the drain end; I_D plateaus and is governed strictly by the gate reverse bias V_{GS}.
- **Summary Findings:**
  - The N-channel JFET functions as a Voltage-Controlled Current Source regulated via V_{GS}.
  - Reverse biasing the Gate-Source junction provides very high input impedance on the order of gigohms.